Tejus Surendran
Pittsburgh, PA | Phone – (978) 427 3884 | Email – tsurendr@andrew.cmu.edu

Biomedical Engineering PhD candidate at Carnegie Mellon University with experience in machine learning, computer vision, medical imaging, and biomechanics. My work spans quantitative imaging, predictive modeling, computational pathology, and vision-guided robotics.

EDUCATION

Carnegie Mellon University
PhD in Biomedical Engineering
Expected Graduation: May 2028
Supervisors: Dr. John Galeotti and Dr. Axel Moore
GPA: 3.95
Applying computer vision, medical image analysis, robotics, and machine learning to quantify cartilage deformation and recovery.

Rensselaer Polytechnic Institute
BS in Computer Science
Degree Conferral: May 2021
Cum Laude
GPA: 3.7
Built computer vision and sensor-guided robotics systems for automated hydrogen-forklift fueling.

EMPLOYMENT

Staff Scientist - Kolachalama Lab – Boston University School of Medicine, Boston, MA (10/2021 – 02/2023)
Developed deep learning and medical imaging methods for imaging-based biomarkers, disease prediction, computational pathology, and brain information flow modeling.

ADDITIONAL RESEARCH & TECHNICAL PROJECTS

RSNA 2026 Knee Abnormality Detection Challenge
Project Focus: LLM-Assisted Medical Image Analysis
Objective: Developing an LLM-assisted medical imaging pipeline that uses large language models to extract structured supervision from multilingual radiology reports. Trained deep visual representations from multi-sequence knee MRI using attention-based aggregation, with report-derived soft labels providing cross-modal supervision for abnormality detection.
Placed : 189 out of 6425 teams

Markerless Human Motion Analysis (Ongoing)
Project Focus: On-Device Computer Vision & Biomechanics
Objective: Develop computer vision and machine learning methods for marker less human-motion analysis from consumer smartphone video, with models designed for local on-device inference and quantitative movement assessment without specialized motion-capture hardware.

RESEARCH EXPERIENCE

Moore Lab – Carnegie Mellon University (08/2023 - Present)
Research Focus 1: Computer Vision & Medical Image Segmentation
Objective: Develop and validate an automated qDESS MRI pipeline for femoral cartilage, tibial cartilage, and meniscus segmentation using transfer learning from larger DESS datasets, followed by serial registration and regional cartilage thickness and T2 analysis.

Research Focus 2: In Vivo Medical Image Analysis
Objective: Quantify task-dependent changes in cartilage thickness, T2 relaxation time, and strain across serial MRI scans following daily activities.
Research Outputs: Presentations (2 poster)

Research Focus 3: Robotic Biomechanics
Objective: Develop a robotically controlled whole-joint model to quantify cartilage deformation and recovery under physiological loading.

Research Outputs: 1 peer-reviewed manuscript, 4 presentations

Kolachalama Lab – Boston University School of Medicine (06/2019 - 02/2023)
Research Focus 1: Medical Imaging & Predictive Modeling
Objective: Develop 2D–3D knee imaging pipelines to quantify subchondral bone geometry and evaluate imaging-derived and clinical biomarkers for predicting total knee replacement
Research Outputs: Peer Reviewed Manuscripts (2)

Research Focus 2: Computational Pathology & Image Analysis
Objective: Build patch-based image-analysis pipelines for whole-slide kidney biopsy classification
Research Output: Presentations (1 poster)

CATS Laboratory - RPI (01/2019 - 05/2020)
Research Focus 1: Computer Vision & Robotics
Objective: Develop vision-guided robotic alignment for automated forklift fueling using AR tag detection
Research Outputs: Implemented on-site

PEER REVIEWED PUBLICATIONS

1. Modeling Physiological Tasks and their Impact on Cartilage Deformation and Recovery: An Intact Porcine Knee Joint Model. **Tejus Surendran**, Axel C. Moore. **Journal of the Mechanical Behavior of Biomedical Materials** 2026

   Significance: Demonstrates that cartilage deformation is time-dependent and task-dependent, with measurable compression during standing and recovery during walking and supine unloading under near-physiologic whole-joint loading.

   Contribution: Develops a repeatable robotic whole-joint porcine knee platform that quantifies cartilage deformation/recovery and detects changes after meniscus/ligament resection, enabling mechanistic studies and intervention testing.

2. Survival analysis on subchondral bone length for total knee replacement. **Tejus Surendran**, Lisa K. Park, Meagan V. Lauber, Baekdong Cha, Ray S. Jhun, Terence D. Capellini, Deepak Kumar, David T. Felson, Vijaya B. Kolachalama. **Skeletal Radiology** 2024

   Significance: Evaluated a deep-learning-derived imaging biomarker against established clinical measures for osteoarthritis prognosis and total knee replacement prediction.

   Contribution: Demonstrated the potential of automated subchondral bone length measurements as a prognostic imaging biomarker for knee osteoarthritis.

3. Subchondral Bone Length in Knee Osteoarthritis: A Deep Learning–Derived Imaging Measure and Its Association with Radiographic and Clinical Outcomes. Gary H. Chang, Lisa K. Park, Nina A. Le, Ray S. Jhun, **Tejus Surendran**, Joseph Lai, Hojoon Seo, Nuwapa Promchotichai, Grace Yoon, Jonathan Scalera, Terence D. Capellini, David T. Felson, Vijaya B. Kolachalama. **Arthritis and Rheumatology** 2021

   Significance: Develop a bone shape measure that reflects the extent of cartilage loss and bone flattening in knee osteoarthritis (OA) and test it against estimates of disease severity.
   Contribution: Our deep learning framework enables automated, large-scale quantification of knee joint structures, enhancing the identification of shape-specific osteoarthritis risk factors and streamlining clinical analysis.

HONORS AND AWARDS

- CMU College of Engineering Presidential Fellowship – 2026
- Outstanding Poster Award – 2025 CMU BME Forum
- Graduated Cum Laude (08/2018 – 05/2021)
- Dean's List Award recipient (08/2018 - 05/2021)
- Academic Excellence Award (01/2020 - 05/2020)

NATIONAL & INTERNATIONAL CONFERENCE PRESENTATIONS

1. Modeling Physiological Tasks and their Impact on Cartilage Deformation and Recovery: A Robotically Controlled Joint Simulator (Poster), **World Congress of Biomechanics**, Vancouver, CA (2026)
2. In Vivo Cartilage Deformation Across a Series of Tasks in Adults with Knee Osteoarthritis (Poster), **ORS Annual**

Meeting, Charlotte, NC (2026)

3. Acute Cartilage Compression and Recovery a Cadaveric Joint Model of Simulated Standing and Walking (Poster), **ORS Annual Meeting**, Charlotte, NC (2026)
4. In-Vivo Joint Dynamics Conserve the Compression and Recovery Response of Cartilage in Cadaveric Joints (Poster), **ASB**, Pittsburgh, PA (2025)
5. Learning to Walk Before You Run: Modeling Gait with a Robotic Simulator (**Invited** Oral Presentation), **ASME SBC/SB3C**, Santa Ana Pueblo, NM (2025)
6. In-Vivo Joint Dynamics Conserve the Compression and Recovery Response of Cartilage in Cadaveric Joints (Oral Presentation), **ASME SBC/SB3C**, Santa Ana Pueblo, NM (2025)
7. Can Walking Recover Cartilage T2 relaxation Time in Healthy and Osteoarthritic Knees (Oral Presentation), **Biomedical Engineering Society**, Baltimore, MD (2024)
8. The Effects of Standing and Walking on Cartilage T2 relaxation time (Poster), **Northeast Bioengineering Conference**, Hoboken, NJ, (2024)